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EXPERIMENT 9

Aim - To implement a Private Ethereum Blockchain using Geth.

Theory -

Ethereum is an open-source, decentralized blockchain platform that enables developers to build and deploy **smart contracts** and **decentralized applications (DApps)**. Unlike public blockchains such as the Ethereum mainnet, a **Private Ethereum Blockchain** is restricted to specific participants, making it ideal for **development, testing, and enterprise applications** where confidentiality, control, and performance are essential.

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through **JSON-RPC, command-line, or JavaScript consoles**.

Setting up a **Private Ethereum Network** involves creating a customized blockchain environment isolated from the public Ethereum network. It enables developers to test smart contracts, consensus mechanisms, and network behavior without incurring gas costs or depending on public validators.

Steps Involved:

1. **Choosing a Network ID:**
2. Each private blockchain must have a unique network ID to differentiate it from public or other private networks.
3. **Choosing a Consensus Algorithm:**
4. Ethereum supports different consensus mechanisms such as **Proof of Work (PoW)** and **Proof of Authority (PoA)**. In private networks, **Clique (PoA)** is often used for faster block confirmation.
5. **Creating a Genesis Block:**
6. The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.
7. **Initializing the Geth Database:**
8. The `geth init` command initializes the blockchain using the `genesis.json` file, preparing the data directory for node operation.
9. **Setting up Networking:**
10. Nodes communicate using **enodes** (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.
11. **Running the Member Nodes:**

12. Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.
13. **Running a Signer (in Clique):**
14. In the **Clique consensus**, designated **signers** validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

Outputs -

1. First install all the necessary dependencies like curl, git, build-essential.
`sudo apt install wget curl git build-essential -y`

```
pranav@DESKTOP-OSHSLJ1:~$ sudo apt install wget curl git build-essential -y
[sudo] password for pranav:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
curl is already the newest version (8.5.0-2ubuntu10.8).
git is already the newest version (1:2.43.0-1ubuntu7.3).
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 11 not upgraded.
```

2. Install geth using wget
`cd /usr/local/bin && sudo wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e501b3b0.tar.gz`

```
pranav@DESKTOP-OSHSLJ1:~$ cd /usr/local/bin && sudo wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e501b3b0.tar.gz
--2026-03-31 13:59:45-- https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e501b3b0.tar.gz
Resolving gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)... 20.60.40.164
Connecting to gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)[20.60.40.164]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 26856444 (26M) [application/octet-stream]
Saving to: 'geth-linux-amd64-1.12.0-e501b3b0.tar.gz.1'

geth-linux-amd64-1.12.0-e501b 100%[=====] 25.61M 66.3KB/s in 2m 40s
2026-03-31 14:02:30 (164 KB/s) - 'geth-linux-amd64-1.12.0-e501b3b0.tar.gz.1' saved [26856444/26856444]
```

3. Extract the tar file and move the geth into /usr/local/bin/geth path & check the geth version
`sudo tar -xvf geth-linux-amd64-1.12.0-e501b3b0.tar.gz`
`sudo mv geth-linux-amd64-1.12.0-e501b3b0/geth /usr/local/bin/geth`
`geth version`

```
pranav@DESKTOP-OSHSLJ1:/usr/local/bin$ sudo tar -xvf geth-linux-amd64-1.12.0-e501b3b0.tar.gz
sudo mv geth-linux-amd64-1.12.0-e501b3b0/geth /usr/local/bin/geth
geth version
geth-linux-amd64-1.12.0-e501b3b0/
geth-linux-amd64-1.12.0-e501b3b0/COPYING
geth-linux-amd64-1.12.0-e501b3b0/geth
Geth
Version: 1.12.0-stable
Git Commit: e501b3b05db8e169f67dc78b7b59bc352b3c638d
Git Commit Date: 20230525
Architecture: amd64
Go Version: go1.20.3
Operating System: linux
GOPATH=
GOROOT=
```

4. Create a folder named private_ethereum_setup and create another 2 directories naming node1 and node2 individually inside private_ethereum_setup directory.
`mkdir private_ethereum_setup`
`cd private_ethereum_setup && mkdir node1 node2`

```
pranav@DESKTOP-OSHSLJI:~$ mkdir private_ethereum_setup
cd private_ethereum_setup && mkdir node1 node2
pranav@DESKTOP-OSHSLJI:~/private_ethereum_setup$ ls
node1  node2
```

5. Check the files

6. Create account for node1

```
geth --datadir "node1" account new
```

```
pranav@DESKTOP-OSHSLJI:~/private_ethereum_setup$ geth --datadir "node1" account new
INFO [03-31|14:05:28.888] Maximum peer count      ETH=50 LES=0 total=50
INFO [03-31|14:05:28.888] Smartcard socket not found, disabling  err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key: 0xC99cf64ef49425Bf0e814c9d51364281a61747d2
Path of the secret key file: node1/keystore/UTC--2026-03-31T14-05-36.977302946Z--c99cf64ef49425bf0e814c9d51364281a61747d2

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

7. Do the same for node2

```
geth --datadir "node2" account new
```

```
pranav@DESKTOP-OSHSLJI:~/private_ethereum_setup$ geth --datadir "node2" account new
INFO [03-31|14:05:46.530] Maximum peer count      ETH=50 LES=0 total=50
INFO [03-31|14:05:46.531] Smartcard socket not found, disabling  err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key: 0x19F1609855626D5551A589497aCb92966c69C778
Path of the secret key file: node2/keystore/UTC--2026-03-31T14-05-58.107593532Z--19f1609855626d5551a589497acb92966c69c778

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

8. Create a file called genesis.json in the same directory and copy paste these contents

```
{
  "config": {
    "chainId": 12345,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "istanbulBlock": 0,
    "berlinBlock": 0,
```



```

GNU nano 7.2                                genesis.json
"config": {
  "chainId": 12345,
  "homesteadBlock": 0,
  "eip150Block": 0,
  "eip155Block": 0,
  "eip158Block": 0,
  "byzantiumBlock": 0,
  "constantinopleBlock": 0,
  "petersburgBlock": 0,
  "istanbulBlock": 0,
  "berlinBlock": 0,
  "londonBlock": 0,
  "clique": {
    "period": 5,
    "epoch": 30000
  }
},
"difficulty": "1",
"gasLimit": "8000000",
"extradata": "0x0000000000000000000000000000000000000000000000000000000000000000c99cf64ef49425bf0e814c9d51364281a61747d200000000000000000000000000000",
"alloc": {
  "0xc99cf64ef49425bf0e814c9d51364281a61747d2": {
    "balance": "1000000000000000000"
  },
  "0x19f1609855626D5551A589497aCb92966c69C778": {
    "balance": "1000000000000000000"
  }
},
"config": {
  "chainId": 12345,
  "homesteadBlock": 0,
  "eip150Block": 0,

```

Read 59 lines

9. Initializing geth for node1 using the genesis.json file

geth init --datadir node1 genesis.json

```

pranav@DESKTOP-USHSLJ1:~/private_ethereum_setup$ geth init --datadir node1 genesis.json
INFO [03-31|14:14:02.677] Maximum peer count                ETH=50 LES=0 total=50
INFO [03-31|14:14:02.677] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [03-31|14:14:02.680] Set global gas cap                  cap=50,000,000
INFO [03-31|14:14:02.680] Initializing the KZG library         backend=gokzg
INFO [03-31|14:14:02.729] Defaulting to pebble as the backing database
INFO [03-31|14:14:02.729] Allocated cache and file handles     database=/home/pranav/private_ethereum_setup/node1/genesis.json
th/chaindata cache=16.00MiB handles=16
INFO [03-31|14:14:02.805] Opened ancient database              database=/home/pranav/private_ethereum_setup/node1/genesis.json
th/chaindata/ancient/chain readonly=false
INFO [03-31|14:14:02.805] Writing custom genesis block
INFO [03-31|14:14:02.805] Persisted trie from memory database  nodes=3 size=411.00B time="33.822µs" gcnodes=0 gcsiz
=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [03-31|14:14:02.826] Successfully wrote genesis state      database=chaindata hash=08f6ea..15522b
INFO [03-31|14:14:02.826] Defaulting to pebble as the backing database
INFO [03-31|14:14:02.826] Allocated cache and file handles     database=/home/pranav/private_ethereum_setup/node1/genesis.json
th/lightchaindata cache=16.00MiB handles=16
INFO [03-31|14:14:02.899] Opened ancient database              database=/home/pranav/private_ethereum_setup/node1/genesis.json
th/lightchaindata/ancient/chain readonly=false
INFO [03-31|14:14:02.899] Writing custom genesis block
INFO [03-31|14:14:02.900] Persisted trie from memory database  nodes=3 size=411.00B time="22.46µs" gcnodes=0 gcsiz
=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [03-31|14:14:02.912] Successfully wrote genesis state      database=lightchaindata hash=08f6ea..15522b

```

10. Do the same for node2

geth init --datadir node2 genesis.json

```

pranav@DESKTOP-OSHSLJ1:~/private_ethereum_setup$ geth init --datadir node2 genesis.json
INFO [03-31|14:14:07.414] Maximum peer count          ETH=50 LES=0 total=50
INFO [03-31|14:14:07.415] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [03-31|14:14:07.418] Set global gas cap          cap=50,000,000
INFO [03-31|14:14:07.418] Initializing the KZG library backend=gokzg
INFO [03-31|14:14:07.448] Defaulting to pebble as the backing database
INFO [03-31|14:14:07.448] Allocated cache and file handles database=/home/pranav/private_ethereum_setup/node2/chaindata cache=16.00MiB handles=16
INFO [03-31|14:14:07.520] Opened ancient database      database=/home/pranav/private_ethereum_setup/node2/chaindata/ancient/chain readonly=false
INFO [03-31|14:14:07.520] Writing custom genesis block
INFO [03-31|14:14:07.520] Persisted trie from memory database nodes=3 size=411.00B time="111.483µs" gcnodes=0 gcsizes=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [03-31|14:14:07.533] Successfully wrote genesis state database=chaindata hash=08f6ea..15522b
INFO [03-31|14:14:07.534] Defaulting to pebble as the backing database
INFO [03-31|14:14:07.534] Allocated cache and file handles database=/home/pranav/private_ethereum_setup/node2/lightchaindata cache=16.00MiB handles=16
INFO [03-31|14:14:07.597] Opened ancient database      database=/home/pranav/private_ethereum_setup/node2/lightchaindata/ancient/chain readonly=false
INFO [03-31|14:14:07.597] Writing custom genesis block
INFO [03-31|14:14:07.598] Persisted trie from memory database nodes=3 size=411.00B time="32.835µs" gcnodes=0 gcsizes=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [03-31|14:14:07.610] Successfully wrote genesis state database=lightchaindata hash=08f6ea..15522b

```

11. Now we will install bootnode for configuring both the nodes.

```

sudo add-apt-repository -y ppa:ethereum/ethereum
sudo apt-get update
sudo apt-get install bootnode

```

12. Create a key for the bootnode and save it to boot.key in the folder

```
bootnode -genkey boot.key
```

```

deilsy@deilsy:~/private_ethereum_setup$ bootnode -genkey boot.key
deilsy@deilsy:~/private_ethereum_setup$ ls
boot.key genesis.json node1 node2
deilsy@deilsy:~/private_ethereum_setup$

```

13. Copy this command and copy the enode://.... This will be required later

```
bootnode -nodekey boot.key -verbosity 9 -addr :30305
```

```

pranav@DESKTOP-OSHSLJ1:~/private_ethereum_setup$ bootnode -nodekey boot.key -addr :30305
enode://65150fcfbf2bac27c639de2d15c6265f8135751d09041a248806606fef4a44206a0bf0673458a5dcb1146bda9aa54d511cbcabfd26c807257191ec1c77b464720127.0.0.1:0?discport=30305
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [03-31|14:15:43.440] New local node record      seq=1,774,966,543 id=c51b56f7e246677f ip="invalid IP" udp=0 tcp=0

```

14. Now on another terminal we will start mining node1

```

geth --datadir node1 --port 30306 --bootnodes
enode://3f690f88c00ce563d9dd609b0674ad97df9fc6b77673bd86917f51333118e81f54e75919746a331d55af7f6edba870a5ef5ed12b60af554ccb922bc77ab41f270127.0.0.1:0?discport=30305 --networkid 123454321
--unlock 0x563A5a670260417B163AD1696fCBfA6849906d06 --password
node1/password.txt --authrpc.port 8551 --miner.etherbase
0x563A5a670260417B163AD1696fCBfA6849906d06 --mine

```

Replace the address with your own created

```

pranav@DESKTOP-0SHSLJ1: /private_ethereum_setup$ geth --datadir node1 \
--port 30306 \
--networkid 12345 \
--bootnodes "enode://65150fcfbf2bac27c639de2d15c265f8135751d09041a24880606fef4a44206a0bf073458a5dcb1146bda9aa54d511bcabfd26c807257191ec1c77b46472@127.0.0.1:30305" \
--mine \
--miner.etherbase 0xC99cf64ef49425Bf0e814c9d51364281a61747d2 \
--http --http.port 8545 --http.api "eth,net,web3,personal,miner" \
--authrpc.port 8551 \
--allow-insecure-unlock \
--unlock 0xC99cf64ef49425Bf0e814c9d51364281a61747d2
INFO [03-31|14:17:28.316] Maximum peer count          ETH=50 LES=0 total=50
INFO [03-31|14:17:28.317] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [03-31|14:17:28.320] Set global gas cap          cap=50,000,000
INFO [03-31|14:17:28.320] Initializing the KZG library backend=gokzg
INFO [03-31|14:17:28.355] Allocated trie memory caches clean=154.00MiB dirty=256.00MiB
INFO [03-31|14:17:28.355] Using pebble as the backing database database=/home/pranav/private_ethereum_setup/node1/geth/chaindata cache=512.00MiB handles=524,288
INFO [03-31|14:17:28.355] Allocated cache and file handles database=/home/pranav/private_ethereum_setup/node1/geth/chaindata/ancient/chain readonly=false
INFO [03-31|14:17:28.395] Opened ancient database
INFO [03-31|14:17:28.396] Initialising Ethereum protocol network=12345 dbversion=<nil>
INFO [03-31|14:17:28.396] -----
INFO [03-31|14:17:28.396] Chain ID: 12345 (unknown)
INFO [03-31|14:17:28.396] Consensus: Clique (proof-of-authority)
INFO [03-31|14:17:28.396] Pre-Merge hard forks (block based):
INFO [03-31|14:17:28.396]   - Homestead: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/homestead.md)
INFO [03-31|14:17:28.396]   - Tangerine Whistle (EIP 150): #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/tangerine-whistle.md)

Unlocking account 0xC99cf64ef49425Bf0e814c9d51364281a61747d2 | Attempt 1/3
Password:
INFO [03-31|14:17:35.536] Unlocked account address=0xC99cf64ef49425Bf0e814c9d51364281a61747d2
INFO [03-31|14:17:35.536] Transaction pool price threshold updated price=0
INFO [03-31|14:17:35.536] Transaction pool price threshold updated price=1,000,000,000
INFO [03-31|14:17:35.536] Commit new sealing work number=1 sealhash=2afec2..f3e4eb uncles=0 txs=0 gas=0 fees=0 elapsed="136.626µs"
INFO [03-31|14:17:35.536] Commit new sealing work number=1 sealhash=2afec2..f3e4eb hash=bc1863..339be8 elapsed="395.7µs"
INFO [03-31|14:17:35.536] Successfully sealed new block number=1 sealhash=2afec2..f3e4eb hash=bc1863..339be8 elapsed="430.403µs"
INFO [03-31|14:17:35.537] "⚡ mined potential block" number=1 hash=bc1863..339be8
INFO [03-31|14:17:35.537] Commit new sealing work number=2 sealhash=973592..5f5a4f uncles=0 txs=0 gas=0 fees=0 elapsed="338.367µs"
INFO [03-31|14:17:35.537] Commit new sealing work number=2 sealhash=973592..5f5a4f uncles=0 txs=0 gas=0 fees=0 elapsed="637.7µs"
INFO [03-31|14:17:39.308] Looking for peers peercount=0 tried=1 static=0
INFO [03-31|14:17:40.001] Successfully sealed new block number=2 sealhash=973592..5f5a4f hash=ae892f..415799 elapsed=4.463s
INFO [03-31|14:17:40.001] "⚡ mined potential block" number=2 hash=ae892f..415799
INFO [03-31|14:17:40.001] Commit new sealing work number=3 sealhash=4e5126..8e1a45 uncles=0 txs=0 gas=0 fees=0 elapsed="358.797µs"
INFO [03-31|14:17:40.001] Commit new sealing work number=3 sealhash=4e5126..8e1a45 uncles=0 txs=0 gas=0 fees=0 elapsed="631.994µs"
INFO [03-31|14:17:45.001] Successfully sealed new block number=3 sealhash=4e5126..8e1a45 hash=ace1d8..8d69c0 elapsed=4.999s
INFO [03-31|14:17:45.001] "⚡ mined potential block" number=3 hash=ace1d8..8d69c0
INFO [03-31|14:17:45.001] Commit new sealing work number=4 sealhash=394b55..a0be7f uncles=0 txs=0 gas=0 fees=0 elapsed="351.696µs"
INFO [03-31|14:17:45.001] Commit new sealing work number=4 sealhash=394b55..a0be7f uncles=0 txs=0 gas=0 fees=0 elapsed="600.929µs"
INFO [03-31|14:17:49.330] Looking for peers peercount=0 tried=0 static=0
INFO [03-31|14:17:50.000] Successfully sealed new block number=4 sealhash=394b55..a0be7f hash=783ee0..2a21ec elapsed=4.999s
INFO [03-31|14:17:50.000] "⚡ mined potential block" number=4 hash=783ee0..2a21ec
INFO [03-31|14:17:50.002] Commit new sealing work number=5 sealhash=0b3313..f3a620 uncles=0 txs=0 gas=0 fees=0 elapsed=1.601ms
INFO [03-31|14:17:50.003] Commit new sealing work number=5 sealhash=0b3313..f3a620 uncles=0 txs=0 gas=0 fees=0 elapsed=2.294ms
INFO [03-31|14:17:55.001] Successfully sealed new block number=5 sealhash=0b3313..f3a620 hash=7fd986..df3592 elapsed=4.999s
INFO [03-31|14:17:55.001] "⚡ mined potential block" number=5 hash=7fd986..df3592
INFO [03-31|14:17:55.002] Commit new sealing work number=6 sealhash=1bdf88..5e952b uncles=0 txs=0 gas=0 fees=0 elapsed="509.252µs"
INFO [03-31|14:17:55.002] Commit new sealing work number=6 sealhash=1bdf88..5e952b uncles=0 txs=0 gas=0 fees=0 elapsed="999.701µs"
INFO [03-31|14:17:59.347] Looking for peers peercount=0 tried=0 static=0
INFO [03-31|14:18:00.000] Successfully sealed new block number=6 sealhash=1bdf88..5e952b hash=d6641b..dd3338 elapsed=4.998s
INFO [03-31|14:18:00.001] "⚡ mined potential block" number=6 hash=d6641b..dd3338
INFO [03-31|14:18:00.002] Commit new sealing work number=7 sealhash=667df4..412fc9 uncles=0 txs=0 gas=0 fees=0 elapsed=1.081ms
INFO [03-31|14:18:00.002] Commit new sealing work number=7 sealhash=667df4..412fc9 uncles=0 txs=0 gas=0 fees=0 elapsed=1.667ms

```

15. Now do same for node2

```

geth --datadir node2 --port 30307 --bootnodes
enode://3f690f88c00ce563d9dd609b0674ad97df9fc6b77673bd86917
f51333118e81f54e75919746a331d55af7f6edba870a5ef5ed12b60af55
4ccb922bc77ab41f27@127.0.0.1:0?discport=30305 --networkid
123454321 --unlock
0x6d04eA66725A40F5669E2d6f4f77eb4Ecf756a3f --password
node2/password.txt --authrpc.port 8552

```

```

pranav@DESKTOP-OSHSLJ1:~/private_ethereum_setup$ geth --datadir node2 \
--port 30307 \
--networkid 12345 \
--bootnodes "enode://65150fcfbf2bac27c639de2d15c6265f8135751d09041a248806606fef4a44206a0bf0673458a5dcb1146bda9aa54d511cbcabfd26c807257191ec1c77b46472@127.0.0.1:30305" \
--http --http.port 8546 --http.api "eth,net,web3" \
--authrpc.port 8552
INFO [03-31|14:18:32.384] Maximum peer count ETH=50 LES=0 total=50
INFO [03-31|14:18:32.385] Smartcard socket not found, disabling err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [03-31|14:18:32.387] Set global gas cap cap=50,000,000
INFO [03-31|14:18:32.387] Initializing the KZG library backend=gokzg
INFO [03-31|14:18:32.433] Allocated trie memory caches clean=154.00MiB dirty=256.00MiB
INFO [03-31|14:18:32.433] Using pebble as the backing database
INFO [03-31|14:18:32.433] Allocated cache and file handles database=/home/pranav/private_ethereum_setup/node2/geth/chaindata cache=512.00MiB handles=524,288
INFO [03-31|14:18:32.479] Opened ancient database database=/home/pranav/private_ethereum_setup/node2/geth/chaindata/ancient/chain readonly=false
INFO [03-31|14:18:32.479] Initialising Ethereum protocol network=12345 dbversion=<nil>
INFO [03-31|14:18:32.480] -----
INFO [03-31|14:18:32.480] Chain ID: 12345 (unknown)
INFO [03-31|14:18:32.480] Consensus: Clique (proof-of-authority)
INFO [03-31|14:18:32.480] Pre-Merge hard forks (block based):
INFO [03-31|14:18:32.480] - Homestead: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/homestead.md)
INFO [03-31|14:18:32.480] - Tangerine Whistle (EIP 150): #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/tangerine-whistle.md)
INFO [03-31|14:18:32.480] - Spurious Dragon/1 (EIP 155): #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/spurious-dragon-1.md)
INFO [03-31|14:18:32.480] - Spurious Dragon/2 (EIP 158): #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/spurious-dragon-2.md)
INFO [03-31|14:18:32.480] - Byzantium: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/byzantium.md)
INFO [03-31|14:18:32.480] - Constantinople: #0 (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrade/constantinople.md)
INFO [03-31|14:18:32.515] Started P2P networking
INFO [03-31|14:18:32.516] IPC endpoint opened url=/home/pranav/private_ethereum_setup/node2/geth.ipc
INFO [03-31|14:18:32.517] Generated JWT secret path=/home/pranav/private_ethereum_setup/node2/geth/jwtsecret
INFO [03-31|14:18:32.517] HTTP server started endpoint=127.0.0.1:8546 auth=false prefix= cors= vhosts=localhost
INFO [03-31|14:18:32.518] WebSocket enabled url=ws://127.0.0.1:8552
INFO [03-31|14:18:32.518] HTTP server started endpoint=127.0.0.1:8552 auth=true prefix= cors=localhost vhosts=localhost
INFO [03-31|14:18:43.349] Block synchronisation started
INFO [03-31|14:18:43.365] Syncing: state download in progress synced=31.92% state=196.00B accounts=1@196.00B slots=0@0.00B codes=0@0.00B eta=14.833ms
INFO [03-31|14:18:43.365] Imported new chain segment number=14 hash=e2f217..cce6a9 blocks=14 txs=0 mgas=0.000 elapsed=1.095ms mgasps=0.000
INFO [03-31|14:18:43.365] Syncing: chain download in progress dirty=0.00B
INFO [03-31|14:18:43.365] Syncing: state download in progress synced=+Inf% chain=18.00B headers=14@6.00B bodies=14@6.00B receipts=14@6.00B eta=-2.290ms
INFO [03-31|14:18:43.365] Syncing: state download in progress synced=43.80% state=392.00B accounts=2@392.00B slots=0@0.00B codes=0@0.00B eta=9.446ms
WARN [03-31|14:18:43.365] Synchronisation failed, retrying err="sync cancelled"
WARN [03-31|14:18:43.366] Unexpected account range packet peer=2d9ec9f1 reqid=4,747,147,368,783,890,389
INFO [03-31|14:18:43.373] Looking for peers peercount=1 tried=2 static=0
WARN [03-31|14:18:45.003] Snap syncing, discarded propagated block number=15 hash=446164..829702
INFO [03-31|14:18:53.368] Syncing: state download in progress synced=43.80% state=392.00B accounts=2@392.00B slots=0@0.00B codes=0@0.00B eta=12.843s
INFO [03-31|14:18:53.370] Syncing: chain download in progress synced=100.00% chain=18.00B headers=16@6.00B bodies=14@6.00B receipts=14@6.00B eta=0s
INFO [03-31|14:18:53.371] Syncing: state download in progress synced=75.10% state=392.00B accounts=2@392.00B slots=0@0.00B codes=0@0.00B eta=3.320s
WARN [03-31|14:18:53.371] Unexpected account range packet peer=2d9ec9f1 reqid=7,480,776,852,441,583,095
INFO [03-31|14:18:53.374] Syncing: state download in progress synced=100.00% state=392.00B accounts=2@392.00B slots=0@0.00B codes=0@0.00B eta=-216ns
INFO [03-31|14:18:53.374] Syncing: state healing in progress accounts=0@0.00B slots=0@0.00B codes=0@0.00B nodes=0@0.00B pending=0
INFO [03-31|14:18:53.385] Rebuilding state snapshot
INFO [03-31|14:18:53.385] Committed new head block number=14 hash=e2f217..cce6a9
INFO [03-31|14:18:53.385] Resuming state snapshot generation root=8ce5ed..df499b accounts=0 slots=0 storage=0.00B dangling=0 elapsed="436.508µs"
INFO [03-31|14:18:53.386] Generated state snapshot accounts=2 slots=0 storage=96.00B dangling=0 elapsed="730.995µs"
INFO [03-31|14:18:53.385] Imported new chain segment number=16 hash=d4eae5..ed7afd blocks=2 txs=0 mgas=0.000 elapsed="300.199µs" mgasps=0.000
INFO [03-31|14:18:53.392] Looking for peers peercount=1 tried=0 static=0

```

16. Now we will interact with the blockchain using javascript console
 geth attach node1/geth.ipc

```

pranav@DESKTOP-OSHSLJ1:~/private_ethereum_setup$ geth attach node2/geth.ipc
Welcome to the Geth JavaScript console!

instance: Geth/v1.12.0-stable-e501b3b0/linux-amd64/go1.20.3
at block: 32 (Tue Mar 31 2026 14:20:10 GMT+0000 (UTC))
datadir: /home/pranav/private_ethereum_setup/node2
modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> net.peerCount
1
>

```

17. Interacting with the blockchain

